

Case Study: Who is the Counter-Strike GOAT?

Executive Summary

Determine the Greatest Counter-Strike Player of All Time (GOAT) by combining CS:GO and CS2 career data (2012–Present) into a weighted formula that evaluates individual statistical performance, trophy achievements, and career longevity. Only players with at least one HLTV Top 20 selection are considered.

Data Collection & Methodology

Dataset Decision:
Sources: HLTV.org + Liquipedia
Collection method: Manual entry into spreadsheets
Player eligibility: ≥1 HLTV Top 20 selection
Time scope: 2012–Present (CS:GO + CS2 combined)
CS 1.6 was initially considered but removed due to insufficient data and the absence of HLTV Top 20 rankings during that era

Key Metrics Established (3 Pillars)

Pillar	Metrics
Pillar 1: Individual Stats	Rating 1.0, K/D Ratio, ADR, KAST%, Impact Rating, MVP Awards Count
Pillar 2: Trophies	Major Championships Won, LAN Wins, LAN Grand Finals Appearances
Pillar 3: Longevity	Years Active, HLTV Ranking Selections (1st through 20th), Total Top 20 Selections, Consistency (years with Rating ≥ 1.05), LANs Played

GOAT Score Formula

Weights: 40% Stats, 30% Trophies, 30% Longevity

Data Preparation

Data Files Collected

File	Pillar	Rows	Columns
Player_Individual_Stats_Dataset_Raw.csv	Pillar 1	98	7 (Player, Rating 1.0, K/D, ADR, KAST, Impact rating, MVPs)
Player_Achievements_Dataset_Raw.csv	Pillar 2	98	4 (Player, Majors won, LANs won, LAN finals)
Player_Longevity_Dataset_Raw.csv	Pillar 3	98	11 (Player, Years active, HLTV 1st–11-20th, HLTV Total, Consistency, LANs played)

Credibility Assessment

Criterion	Rating	Justification
Reliable	✓ High	HLTV is the industry-standard statistics provider
Original	✓ High	Data collected directly from match servers
Comprehensive	✓ High	Covers all players with ≥1 Top 20 selection
Current	✓ High	Includes CS2 era data up to 2026

Cited	✓ High	Universally referenced by teams, analysts, and media
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Limitations & Scope

- CS 1.6 excluded (insufficient data, no Top 20 rankings)
- Some players with very few LANs (e.g., Nico: 16, syrsoN: 19) — small sample sizes flagged
- Rating 1.0 does not capture leadership, utility usage, or clutch pressure

Data Processing

Quality Control & Validation

Check	Result
Row count consistency (98 across all files)	✓ Pass
Missing values	✓ None found
Duplicate players	✓ None found
Player name consistency across files	✓ All match
HLTV Total = sum of ranking columns	✓ Validated

Data Cleaning & Issue Resolution

Issue	File	Fix Applied
KAST stored as text with "%" (e.g., "75.8%")	Individual Stats	Removed "%", converted to numeric
"Years active" stored as text range (e.g., "2016 - Present")	Longevity	Parsed into start_year and end_year; "Present" = 2026
Calculated years_active as	Longevity	end_year - start_year + 1

numeric		
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Technology Stack

Tool	Role	Justification
Spreadsheets	Data cleaning, transformation, GOAT formula calculations	Best for manual data, quick transformations, formula-based scoring
SQL	Store combined dataset, JOINS, aggregations, trend queries	Efficient for merging tables and running summary statistics
Tableau	Visualizations and final presentation	Best for interactive dashboards

Transformation Log

Transformation	Before	After
KAST	"75.8%"	75.8 (numeric)
Years active	"2016 - Present"	start_year: 2016, end_year: 2026, years_active: 11
Column names	Mixed case with spaces	snake_case for SQL compatibility
Weighted Rankings	Separate columns	$(\text{HLTV_1st} \times 6) + (\text{HLTV_2nd} \times 5) + (\text{HLTV_3rd} \times 4) + (\text{HLTV_4_5th} \times 3) + (\text{HLTV_6_10th} \times 2) + (\text{HLTV_11_20th} \times 1)$

Players Flagged (Low Sample Size)

Player	LANs Played
Nico	16
syrsoN	19

swag	19
molodoy	20

Flagged for awareness but NOT removed — they meet eligibility criteria.

Analysis & Results

Step 1: Data Aggregation

Merged 3 tables into one master table (98 rows × 20 columns) using Player as the JOIN key using SQL

Step 2: Normalization Strategy

All metrics normalized to 0–1 range: $\text{normalized_value} = (\text{value} - \text{min}) / (\text{max} - \text{min})$

Key Min/Max Values:

Metric	Min	Max
Rating 1.0	0.95	1.27
K/D	0.95	1.44
ADR	66.1	91.1
KAST%	66.1	76.3
Impact Rating	0.88	1.49
MVPs	0	32
Majors Won	0	5
LANs Won	0	37
LAN Finals	0	50
Years Active	3	14
HLTV Total	1	10
Consistency	0	14

LANs Played	16	141
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Step 3: Calculating the GOAT Score

Pillar 1 Score = Average of normalized: Rating, K/D, ADR, KAST, Impact, MVPs

Pillar 2 Score = Average of normalized: Majors, LANs Won, LAN Finals

Pillar 3 Score = Average of normalized: Years Active, HLTV Total, Consistency, LANs Played, Weighted Rankings

GOAT Score = (Pillar 1 × 0.40) + (Pillar 2 × 0.30) + (Pillar 3 × 0.30)

Step 4: Scale Standardization

All scores multiplied by 100 and rounded to 1 decimal place for readability.

Step 5: Final Rankings (Top 10)

Rank	Player	Stats	Trophies	Longevity	GOAT Score
1	ZywOo	93.9	65.2	68.2	77.6
2	device	58.2	77.9	90.6	73.8
3	s1mple	76.3	34.4	80.1	64.9
4	NiKo	55.8	48.2	92.0	64.4
5	ropz	46.9	75.0	67.5	61.5
6	dupreeh	34.9	87.7	65.6	59.9
7	apEX	18.4	93.3	48.5	49.9
8	Magisk	36.8	69.2	47.5	49.7
9	f0rest	38.2	54.1	52.3	47.2
10	sh1ro	65.1	28.6	41.7	47.1

Step 6: Sensitivity Analysis

Scenario	Weights	#1 Player
Original	40/30/30	ZywOo (77.6)
Equal	33/33/34	device (75.7)
Stats Heavy	50/25/25	ZywOo (80.3)
Trophies Heavy	25/50/25	device (76.1)
Longevity Heavy	25/25/50	device (79.3)

Key Finding: ZywOo is #1 only under original and stats-heavy weights. Device wins under all other scenarios.

Step 7: Key Trends & Qualitative Insights

#	Insight
1	ZywOo dominates individual stats (highest rating, most MVPs, highest impact)
2	device dominates longevity (14 consistency years, 133 LANs, 9 Top 20s)
3	The IGL Paradox: apEX (0.98 rating) and gla1ve (0.96 rating) each won 4 Majors
4	NiKo's curse: 10 Top 20 selections but never ranked #1
5	High rating ≠ MVPs: 8 players with 1.10+ rating but only 1 MVP
6	donk is the most weight-sensitive player (#7 stats-heavy → #35 longevity-heavy)
7	XANTARES matches device's consistency (14 years) but has 0 Majors
8	Short peak vs. sustained excellence: coldzera (67% peak-to-total ratio) vs. s1mple (38%)

Visual Representations

Top 10 GOAT Score

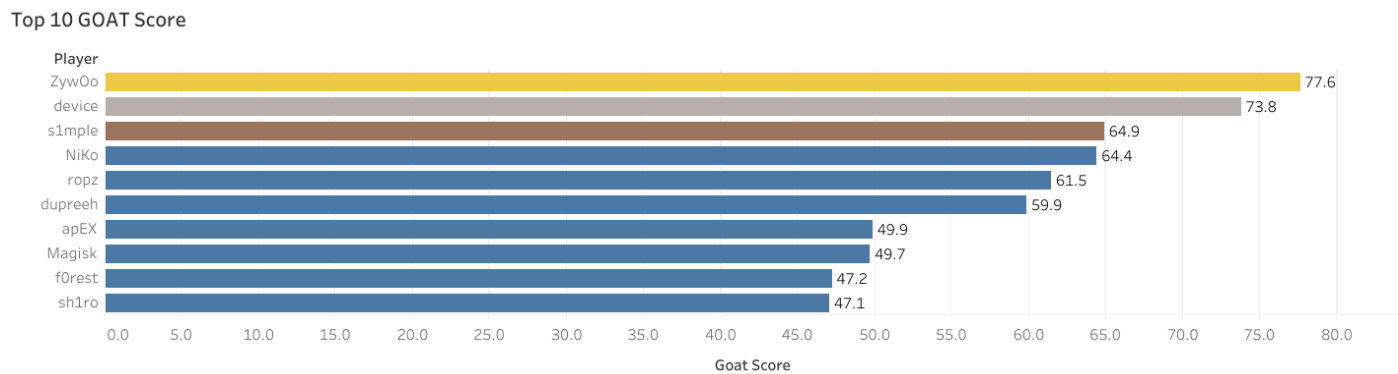


Figure 1: Comparison of the top 10 players based on their overall GOAT Score, with Zywoo currently leading the rankings.

Pillar contribution breakdown

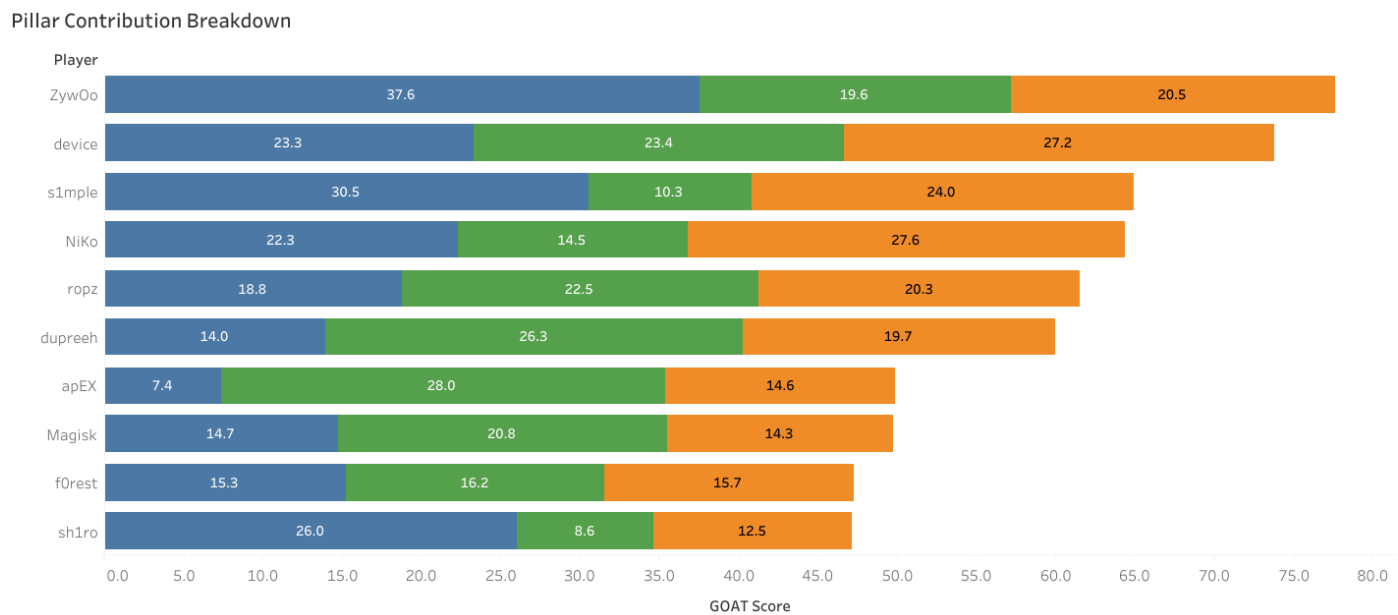
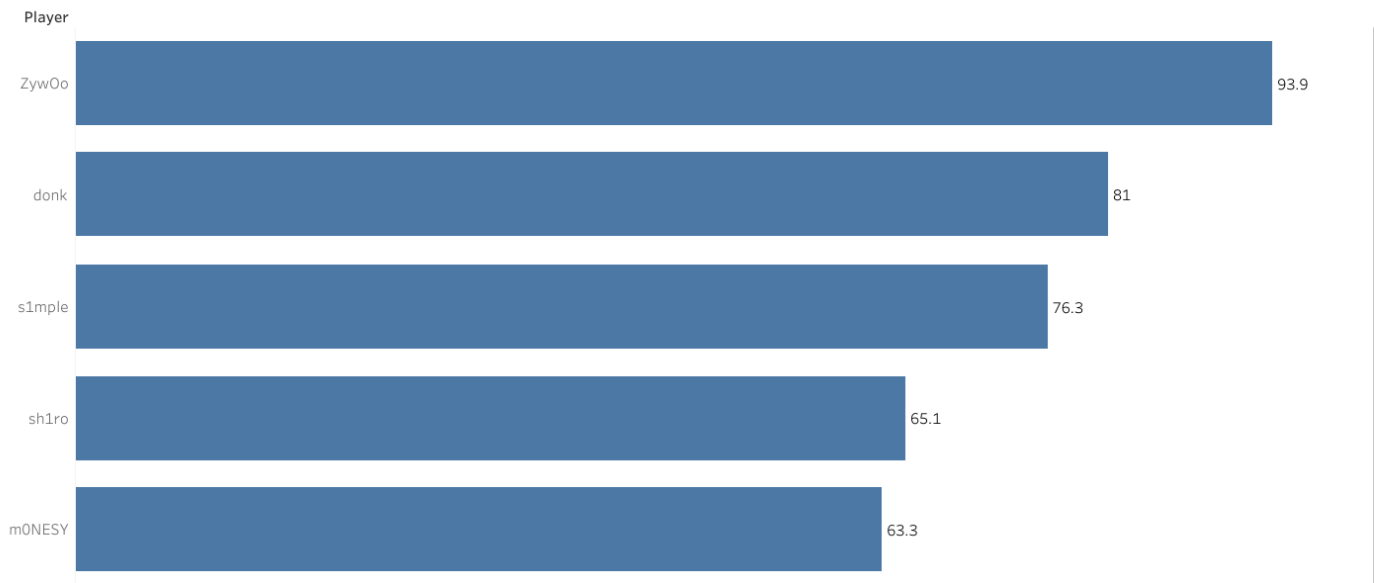


Figure 2: A breakdown of how individual statistical performance, trophy count, and career longevity contribute to each player's final GOAT score.

Pillar Performance Leaders

Top 5 in Stats



Top 5 in Trophies



Top 5 in Longevity

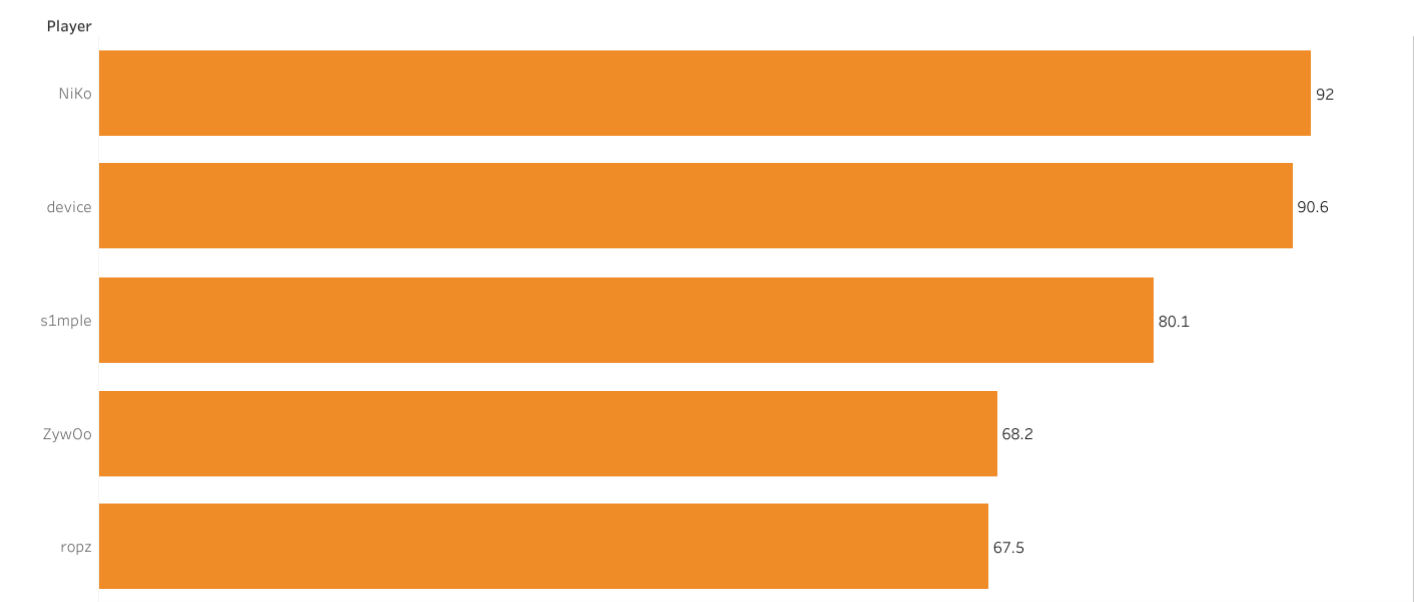


Figure 3: Highlighting the top 5 players based specifically on individual statistical performance metrics such as K/D and ADR.

Figure 4: A ranking of the top 5 players based on their total trophy achievements and major tournament wins.

Figure 5: Highlighting the top 5 players based on career longevity metrics, including years active and consistency ratings.

Sensitivity Analysis

Ranking Comparison Across Weight Scenarios

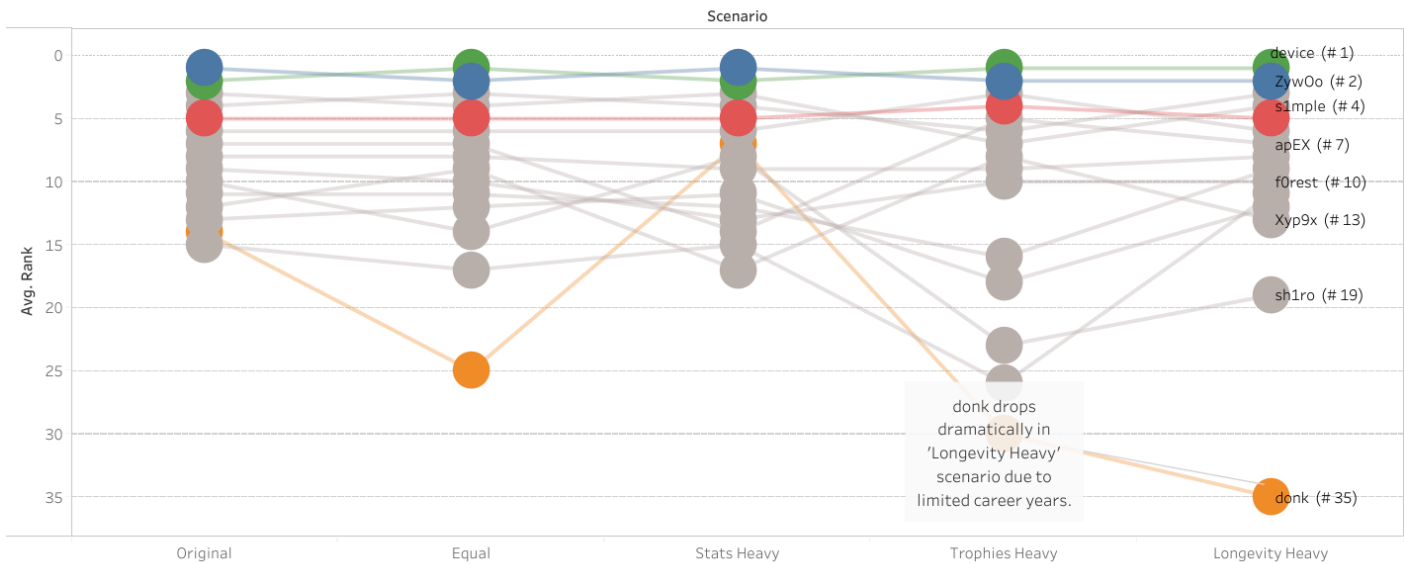


Figure 6: A sensitivity analysis demonstrating how different weight scenarios, such as prioritizing stats versus longevity, shift individual player rankings.

Most Top 20 Without #1

Most Top 20 Selections Without Ever Being #1

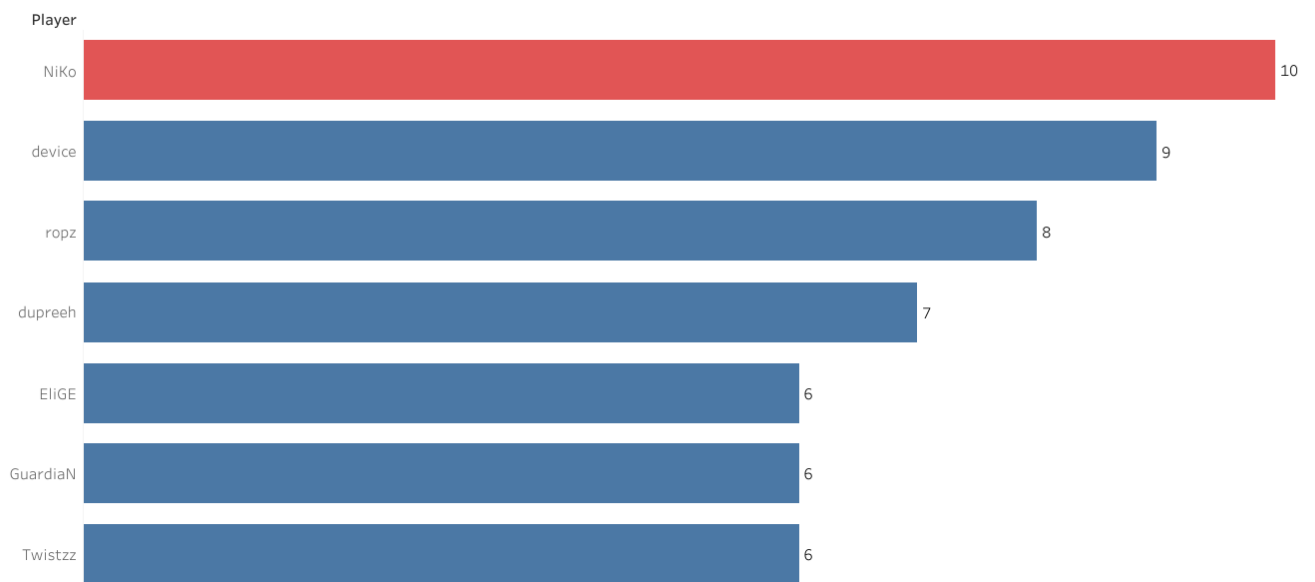


Figure 7: A visualization of players with the most HLTV Top 20 selections who have consistently performed at a high level without achieving the #1 ranking.

Explore the full interactive dashboard here: [Counter Strike GOAT Analysis Dashboard](#)

Conclusion & Key Takeaways

#	Insight
1	ZywOo is the GOAT under the chosen formula (77.6/100)
2	device is the closest challenger and wins under most weight scenarios
3	The GOAT answer depends on what you value

4	s1mple is the most balanced player
5	The IGL Paradox: apEX and gla1ve won 4 Majors with <1.00 rating
6	NiKo's curse: 10 Top 20 selections, never #1
7	donk is the most weight-sensitive player

Future Research & Deliverables

#	Deliverable	Priority
1	Round-level clutch analysis	High
2	Opponent strength adjustment	High
3	Era-separated analysis (CS:GO vs CS2)	Medium
4	Peer voting / expert panel hybrid	Medium
5	Team success correlation	Medium
6	Career trajectory visualization	Medium
7	CS 1.6 inclusion (if data improves)	Low
8	Expanded longevity metrics	Low

Study Limitations

- CS 1.6 excluded due to insufficient data and absence of HLTV Top 20 rankings
- Rating 1.0 does not capture leadership, in-game calling, utility usage, or clutch pressure
- Small sample sizes for some players (e.g., Nico: 16 LANs, syrs0N: 19 LANs)
- Manual data entry introduces potential for human error (mitigated by cross-checking with HLTV)
- Weight selection (40/30/30) is subjective; sensitivity analysis provided to show impact of different weights
- MVP voting bias may favor players on winning teams or more visible roles

Final Statement

This case study demonstrates a complete end-to-end data analysis workflow applied to a subjective esports debate. By structuring the "GOAT" question into three quantifiable pillars and applying a transparent weighted formula, the analysis provides a data-driven framework for evaluating player greatness while acknowledging the inherent limitations of statistics in capturing the full picture of competitive excellence.

The final answer: Under the chosen formula, ZywOo is the GOAT. But the "true" GOAT depends on what you value most — and the data framework lets anyone decide for themselves.